

Janarthanan S

Vellore, Tamil Nadu

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Summary

Python Developer with hands-on experience in Machine Learning, Natural Language Processing, and Computer Vision. Skilled in Object-Oriented Programming, Data Structures & Algorithms, and REST API development. Built real-world AI systems and data-driven applications using Python, OpenCV, and Streamlit. Solved **450+ DSA problems** on LeetCode. Seeking software engineering or data/ML roles to apply strong problem-solving and analytical skills.

Technical Skills

- **Languages:** Python, SQL, HTML, CSS
- **Libraries & Frameworks:** Pandas, NumPy, Scikit-learn, Matplotlib, OpenCV, YOLOv11, Streamlit, Firebase
- **Tools & Concepts:** Git, GitHub, VS Code, Google Cloud | ML, NLP, OOP, DSA, REST API, Debugging

Internship Experience

Proxenix Software Company

July 2025 – September 2025

Machine Learning Intern (Remote)

- Built an NLP sentiment analysis model in Python to decode user behavior patterns, delivering data-backed business insights.
- Managed ML pipeline end-to-end — preprocessing, feature engineering, model evaluation — ensuring high accuracy and on-time delivery.
- Collaborated with cross-functional team to document findings and present data-driven recommendations, improving overall project delivery efficiency.

Projects

Pothole Detection & Real-Time Alert System | Python, YOLOv11, OpenCV, Computer Vision

- Engineered a real-time computer vision pipeline using YOLOv11 object detection and OpenCV with radar sensor fusion, achieving **92% precision** on live road data.
- Implemented automated voice alert system for instant hazard notification; applied deep learning model optimization to improve inference speed and reduce false positives.
- Trained YOLOv11 model on custom-annotated road image dataset; validated system in real-world conditions with consistent and reliable detection performance.

Smart Hostel Issue Tracking System | Python, Streamlit, SQL, Data Analysis

- Designed and deployed a full-stack web application using Streamlit and SQL to centralize student grievance management, reducing issue resolution time significantly.
- Performed data analysis on complaint records to identify recurring infrastructure patterns, enabling data-driven decision making for administration.
- Built admin dashboard to monitor, categorize and resolve student complaints in real time; supported multiple complaint types including electrical, plumbing and maintenance issues.

Personal Finance & Analytics Dashboard | Python, Pandas, NumPy, Matplotlib, Streamlit

- Built an interactive data analytics dashboard to visualize and track spending behavior using Pandas and Matplotlib for data processing and visualization.
- Optimized query logic and data structures achieving **30% faster query performance**, improving real-time dashboard responsiveness.

Aljust Spoken English Platform | AI Voice Agents, Firebase, NLP, Web Development

- Developed an AI-powered interactive language learning platform integrating NLP-based voice agents and Firebase real-time database for dynamic content delivery.
- Improved user engagement by **40%** through localized UX design and personalized learning flows tailored for regional users.

Education

Madras Institute of Technology, Anna University

Sep 2023 – May 2026

B.E. Electronics and Communication Engineering

Chennai, Tamil Nadu

Achievements & Certifications

- Solved **450+ DSA problems** on LeetCode (Python) — strong algorithmic problem-solving skills.
- Grew **LinkedIn community to 2,000+ followers** sharing consistent Python & ML build-in-public content.
- **HackerRank Certifications:** Python (Basic) – Feb 2025 | SQL (Basic) | Problem Solving (Basic) | REST API (Intermediate)
- **Python for Beginners** – Scaler Academy | **Basics of Python** – Udemy

Leadership

- Led cross-functional project teams of 4–6 members in developing software solutions, managing task allocation, sprint planning, and on-time delivery using agile methodology.
- Mentored peers in Python programming and DSA problem-solving techniques, helping team members improve coding efficiency and algorithmic thinking.
- Completed NSO activities demonstrating discipline, teamwork, and adaptability under pressure in a structured environment.